🎬 Demo Scenarios — HR Workflow Designer

> A guide to showcasing the full capability of the HR Workflow Designer across **5 real-world HR use cases.**

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Scenario 1 — Employee Onboarding *(Best for main demo)*

> **Story:** A new software engineer joins the company. HR needs to coordinate across 4 departments before Day 1.

🧩 Nodes to Build

#	Node Type	Title	Key Configuration
1	🟢 **Start**	`New Hire Accepted Offer`	—
2	🔵 **Task**	`Collect Documents`	Assignee: `HR Team` · Due: *3 days before joining*
3	🟡 **Approval**	`Background Verification Clearance`	Approver Role: `HRBP` · Auto-approve threshold: `5 days`
4	🟣 **Automated**	`Send Welcome Email`	Action: `send_email` · `to = new hire` · `subject = "Welcome to the team!"`
5	🟣 **Automated**	`Generate Offer Letter PDF`	Action: `generate_doc` · `template = offer_letter`
6	🔵 **Task**	`Laptop & Access Setup`	Assignee: `IT Team`
7	🟡 **Approval**	`Final HR Sign-off`	Approver Role: `Manager`
8	🔴 **End**	`Onboarding Complete`	Summary flag: `ON`

✅ What It Demonstrates

- **All 5 node types** used in a single workflow
- **Chained approvals** — two approval gates at different stages
- **Automated actions** with dynamic params (`send_email`, `generate_doc`)
- **Full simulation flow** end-to-end through the Sandbox panel

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Scenario 2 — Leave Approval Request

> **Story:** An employee applies for 5 days of annual leave. It needs manager and HR sign-off above a threshold.

🧩 Nodes to Build

#	Node Type	Title	Key Configuration
1	🟢 **Start**	`Leave Request Submitted`	Metadata: `type = Annual Leave`
2	🟡 **Approval**	`Line Manager Approval`	Approver: `Manager` · Auto-approve threshold: `2 days`
3	🟡 **Approval**	`HR Policy Check`	Approver: `HRBP` · Auto-approve threshold: `0` *(manual always)*
4	🟣 **Automated**	`Update Leave Balance`	Action: `send_email` · `to = Payroll` · `subject = "Leave approved"`
5	🔴 **End**	`Leave Confirmed`	End message: `"Employee leave registered successfully"`

✅ What It Demonstrates

- **Sequential approvals** flowing one into the next
- **Auto-approve threshold** behaviour (threshold `0` = always manual)
- **Short, tight workflow** — great for showing the sandbox simulation log quickly

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Scenario 3 — Performance Review Cycle

> **Story:** Annual performance reviews require self-assessment, manager scoring, and HR archiving.

🧩 Nodes to Build

#	Node Type	Title	Key Configuration
1	🟢 **Start**	`Review Cycle Initiated`	Metadata: `quarter = Q4` · `year = 2025`
2	🔵 **Task**	`Employee Self-Assessment`	Assignee: `Employee` · Due: `Nov 15`
3	🔵 **Task**	`Manager Evaluation`	Assignee: `Reporting Manager` · Due: `Nov 30`
4	🟡 **Approval**	`Director Final Review`	Approver: `Director` · Auto-approve threshold: `3`
5	🟣 **Automated**	`Archive Review Document`	Action: `generate_doc` · `template = perf_review` · `recipient = HR Archive`
6	🔴 **End**	`Review Cycle Complete`	Summary flag: `ON`

✅ What It Demonstrates

- **Key-value metadata** on the Start node (`quarter`, `year`)
- **Multiple Task nodes** with different assignees and due dates
- **Document generation** automation at the end of a cycle

Scenario 4 — Employee Offboarding

> **Story:** An employee resigns. HR must revoke access, retrieve assets, and issue clearance.

🧩 Nodes to Build

#	Node Type	Title	Key Configuration
1	🟢 **Start**	`Resignation Received`	Metadata: `last_day = Dec 31`
2	🔵 **Task**	`Exit Interview`	Assignee: `HR Team`
3	🟣 **Automated**	`Revoke System Access`	Action: `send_email` · `to = IT Admin` · `subject = "Access revocation request"`

#	Node Type	Title	Key Configuration
4	🔵 **Task**	`Asset Retrieval`	Assignee: `IT Team` · Custom field: `items = Laptop, ID Card`
5	🟡 **Approval**	`Finance Clearance`	Approver: `Manager` · Auto-approve threshold: `1`
6	🟣 **Automated**	`Generate Experience Letter`	Action: `generate_doc` · `template = exp_letter` · `recipient = employee`
7	🔴 **End**	`Offboarding Complete`	End message: `"All clearances done"`

✅ What It Demonstrates

- **Real-world multi-department coordination** (HR, IT, Finance)
- **Custom key-value fields** on Task nodes (`items = Laptop, ID Card`)
- **Mix of automated + human steps** interleaved throughout the flow

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Scenario 5 — Validation & Error Handling *(Power Move)*

> **Story:** Intentionally build a **broken** workflow to show the sandbox validator catches issues before execution.

🧩 What To Do

Step 1 — Orphan nodes:

> Drop a **Task** node and an **Approval** node onto the canvas with **no connections** between them.

Step 2 — Incomplete path:

> Drop a **Start** node connected *directly* to an **End** node, bypassing all intermediate steps.

Step 3 — Create a cycle:

> Connect `Node A → Node B → Node A` to form a circular dependency.

Step 4 — Hit "Test Workflow" in the Sandbox panel and observe the validation output.

✅ What It Demonstrates

Validation Check	Error Type Caught
Unconnected nodes	Orphan node detection
No intermediate steps	Incomplete path warning

Validation Check	Error Type Caught
Circular connections	Cycle detection in `graphValidator.ts`

> 💡 **Why this matters:** This shows **architectural depth**, not just UI polish. Evaluators specifically look for candidates who handle failure cases — not just the happy path.

🗺️ Recommended Demo Order

```
Scenario 5 (30 sec)  →  Show validator catching a bad graph first
        ↓
Scenario 1 (main)    →  Build the full Onboarding flow from scratch
```

> **Why this order?** Handling failure cases *before* the happy path signals senior engineering instincts to the evaluator. It demonstrates that you think about edge cases as a first-class concern — not an afterthought.

🎨 Node Type Legend

Icon	Node Type	Purpose
🟢	**Start**	Workflow entry point
🔵	**Task**	Human task — collect, interview, setup
🟡	**Approval**	Manager / HR / Director sign-off gate
🟣	**Automated**	System-triggered action (email, doc generation)
🔴	**End**	Workflow completion